On the Puzzles in Cosmic-Ray Observations Underground.

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In this note, we shall describe and discuss two puzzles to be resolved in the cosmic-ray observations underground. The first one, which has been with us for the past few years, is the discrepancy in the zenith-angle distribution of muons between the experiment of BERGESON *et al.* [1] and those of KRISHNASWAMY *et al.* [2] and STOCKEL [3]. If the high-energy muons result from the decays of pions and kaons produced in the atmosphere, one expects [4] an increase with zenith angle in the number of muons above a given energy. While BERGESON *et al.* do not find such an increase, at least by the expected magnitude, KRISHNASWAMY *et al.* and STOCKEL do. The second puzzle, which has come up only recently, is the observation of BASCHIERA *et al.* [5] on the stopping muons underground. These authors find an excess of slow muons that stop and decay in their apparatus, over and above what one would expect from a differential of depth-intensity curve and from assumed/known interactions of muons and neutrinos.

KEUFFEL [6] has proposed a *single* model to explain *both* these puzzles. According to his model, high-energy muons initiate hadronic cascades in the rock in the (100÷1000) GeV range the end products (pions and/or muons with energies $\sim$ few hundred MeV) of which may propagate at very wide angles with respect to the initiating muon. It is these wide-angle, low-energy particles that stop and decay in the apparatus of BASCHIERA *et al.* to account for the observed excess. Likewise KRISHNASWAMY *et al.* and STOCKEL would detect such a flux of low-energy particles, the threshold to trigger their apparatus being only a few hundred MeV, and mistake them for the high-energy muons that were produced in the atmosphere, penetrated all the way down from the

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(1) H. E. BERGESON, J. W. KEUFFEL, M. O. LARSON, E. R. MARTIN and G. M. MASON: *Phys. Rev. Lett.*, **19**, 1487 (1967); see also H. E. BERGESON, J. W. KEUFFEL, M. O. LARSON, G. W. MASON and J. L. OSBORNE: *Phys. Rev. Lett.*, **21**, 1089 (1968); J. W. KEUFFEL *et al.*: *Proceedings of the XI International Conference on Cosmic Rays* (Budapest, 1969), paper MU 31.

(2) M. R. KRISHNASWAMY, M. G. K. MENON, V. S. NARASIMHAM, S. KAWAKAMI, S. KINO and S. MIYAKE: *Phys. Lett.*, **27** B, 535 (1968); *Proceedings of the International Conference on Cosmic Rays* (Budapest, 1969), paper MU 36.

(3) C. T. STOCKEL: *Journ. Phys. A*, **2**, 639 (1969).

(4) See, for example, P. H. BARRETT, L. M. BOLLINGER, G. COCCONI, Y. EISENBERG and K. GREISEN: *Rev. Mod. Phys.*, **24**, 133 (1952).

(5) B. BASCHIERA, L. BERGAMASCO, C. CASTAGNOLI and P. PICCHI: *Lett. Nuovo Cimento*, **4**, 121 (1970); *Lett. Nuovo Cimento* (1971), in press.

(6) J. W. KEUFFEL: *Muon decays underground and Utah effect*, to be published.

surface and traversed the apparatus at a large zenith angle. On the other hand, BERGESON *et al.* do not register them at all, as the threshold in their apparatus is 2 GeV and the particle has to penetrate several interaction mean free paths. Of course, this explanation still leaves the mode of production of high-energy muons in the atmosphere to be understood; but that is a separate question. The puzzle of the discrepancy between the experimental results is resolved, according to the Keuffel model.

In this letter we shall comment on the magnitude of the discrepancy in the second puzzle and present arguments to show that the Keuffel model is probably implausible.

In Fig. 1 the results of BASCHIERA *et al.* on stopping-muon flux are presented along with those of others. There are some modifications and additions in this Figure, compared with a similar one given by BASCHIERA *et al.* The ordinate gives the ratio R

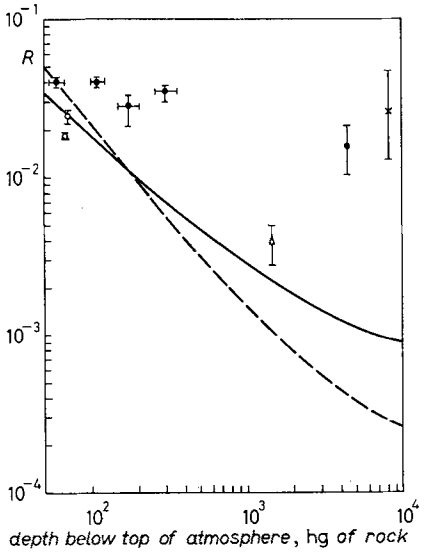


Fig. 1. – Variation of the ratio R of the muon flux stopping in 100 g cm^{-2} to penetrating muon flux as a function of the depth below the top of the atmosphere. The points are from the various experiments: • BASCHIERA *et al.*, ○ BARTON *et al.*, ⊛ SHORT, × REINES, △ KROPP *et al.* The curves are the expected variations, the dashed curve by BASCHIERA *et al.* and the continuous curve by us. R=muons cm^{-2} s sr^{-1} (100 g cm^{-2})$^{-1}$/muons cm^{-2} s sr^{-1}.

of muon flux stopping in 100 g cm^{-2} of matter (BASCHIERA *et al.* give the flux stopping in ~ 75 g cm^{-2}) to the penetrating muon flux. The results of BARTON *et al.* [7] and of SHORT [8] are also shown here. The curve showing the variation of R with depth h as expected by us is shown in addition to that expected by BASCHIERA *et al.* To deduce this curve, BASCHIERA *et al.* start with the energy spectrum of muons at the surface (which was itself deduced largely from the depth-intensity curve) and, using a Monte Carlo program to take into account the fluctuations in energy loss of muons, deduce the expected stopping-muon fluxes. Instead of going through the propagation of errors twice, we have directly deduced our curve by differentiating the known depth-

(7) J. C. BARTON, J. P. BETTS and C. M. POURGOURIDES: *Proceedings of the VI Inter-American Seminar on Cosmic Rays* (La Paz, 1970).

(8) A. M. SHORT: *Proc. Phys. Soc.*, **81**, 541 (1963).

intensity curve (9). The points which we wish to make are the following: i) BASCHIERA *et al.* observe a stopping-muon flux that is greater than the expected one even at shallow ((100÷300) hg; 1 hg = 100 g.cm^{-2} of matter) depths. ii) The result of KROPP *et al.* (10) agrees with the expectation within 1.5 standard deviations. iii) The point attributed to REINES (11) cannot really be considered along with the others. For, at such great depths, the neutrino-produced muon flux, being $\sim 10^{-2}$ of the atmospheric muon flux I_ν, cannot certainly be neglected (*) when compared with the stopping-muon flux which is expected to be $\sim 10^{-3} I_\nu$. We can do so at all other depths shown in the Figure. iv) The flux given by BASCHIERA *et al.* at 4400 hg *is* an order of magnitude higher than the expected one. While this discrepancy seems very real, it appears to the author that there are no other observations showing a similar discrepancy in a convincing manner.

We will first consider the discrepancy in the experiment of BASCHIERA *et al.* at a shallow depth, say 110 hg. The discrepancy in the observed stopping-muon flux, I(slow), is $(2.2 \pm 0.3)\cdot 10^{-2}$ of the vertical muon intensity of 110 hg. I(slow), therefore, is $\sim 5\cdot 10^{-6}$ particles s^{-1} sr^{-1}(100 g)$^{-1}$. It may be pointed out that the average energy of muons at that depth is only ~ 25 GeV. We will now show how this discrepancy cannot be understood in terms of either the known interaction characteristics of muons of energy $\leqslant 25$ GeV or the hypothesized interaction characteristics of muons of energy $\geqslant 300$ GeV, as proposed by KEUFFEL. Several observations were made in the past on the nuclear interactions produced by cosmic-ray muons at shallow depths, notably by the Japanese groups (12). They find that the meson-producing cross-section for muons at shallow depth is $\leqslant 3$ μb/nucleon. Assuming that i) the bulk of the produced mesons have energies $\leqslant 500$ MeV in order to stop in the apparatus, ii) the charged-pion multiplicity is ~ 4 and iii) the effective target thickness is ~ 100 g cm^{-2}, one obtains

$$I(\text{slow}) \approx I_\nu(110\text{ hg}) \times 3\ \mu\text{b} \times 4 \times 100 \times \text{Avogadro number} \approx 1.6\cdot 10^{-7}\ \text{particles/s sr (100 g)}\,,$$

which is lower than the required I(slow) by a factor ~ 30. One needs, to explain the discrepancy, an « anomalous » (hitherto unobserved) $\sigma_{\mu\mathcal{N}} \approx 100$ μb, as already noted by BASCHIERA *et al.*

To see if Keuffel's model can account for the discrepancy, let us (in conformity with the model) assume that i) the total multiplicity of charged pions in the hadronic cascades initiated by muons of energy $\geqslant 300$ GeV is 70 and ii) the « effective » target thickness ~ 300 g·cm^{-2}. One can write down

$$I(\text{slow}) \approx I_\mu(>300\text{ GeV}) \times \sigma_{\mu\mathcal{N}} \times 70 \times 300 \times 6 \times 10^{23}\,.$$

To account for the discrepancy, one requires

$$\sigma_{\mu\mathcal{N}} \approx 5.10^{-6}/(10^{-6} \times 70 \times 300 \times 6 \times 10^{23}) \approx 4\cdot 10^{-28}\ \text{cm}^2\text{/nucleon}\,.$$

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(9) See, for example, the Miyake empirical relation with the constants given by M. G. K. MENON and P. V. RAMANA MURTHY: *Progress in Elementary Particle and Cosmic-Ray Physics*, edited by J. G. WILSON and S. A. WOUTHUYSEN, Vol. **9**, Chapt. III (Amsterdam, 1967), p. 191.

(10) W. R. KROPP, F. REINES and R. M. WOODS jr.: *Phys. Rev. Lett.*, **20**, 1451 (1968).

(11) F. REINES: *Neutrino Meeting*, CERN 69-28 (1969), as quoted in ref. (5); see also ref. (14).

(*) Indeed, REINES *et al.* themselves interpret the muon decays observed by them as due to neutrino-produced muons; see ref. (14).

(12) See, for example, S. HIGASHI, M. ODA, T. OSHIO, H. SHIBATA, K. WATANABE and Y. WATASE: *Progr. Theor. Phys.*, **16**, 250 (1956).

This value for the required cross-section is too large to have passed unnoticed in the cloud chamber experiments underground. Keuffel's model, therefore, does not seem to be successful in explaining the discrepancy at shallow depths.

Let us examine the predictions of the Keuffel's model for the fluxes I_T of particles with relatively low energies capable of penetrating thin detectors at very great depths. In the Table below are shown the relevant details at the two sites at which observations were reported ([13,14]) and predictions are made for I_T, by KEUFFEL. The KGF and ERPM observations on neutrino-produced muon fluxes have to be corrected for I_T by the respective amounts, for both the experiments used thin detectors. When so corrected, the neutrino-produced muon flux values $I_\nu(\mu)$ no longer agree with each other as can be seen from the last row in the Table. The probability that both of them represent a single (average) value of $1.65 \cdot 10^{-13}$ is $6.5 \cdot 10^{-5}$. To find out the maximum value of I_T that is compatible with the ERPM and KGF experiments, we have plotted in Fig. 2 the $I_\nu(\mu)$, corrected for I_T, from each of the two experiments as a function

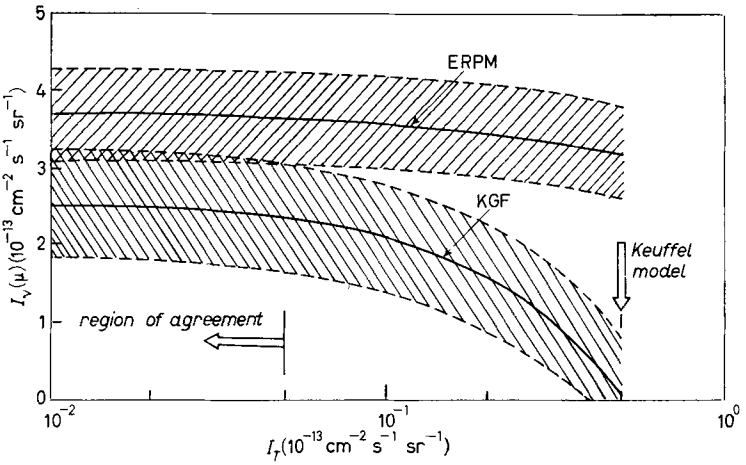


Fig. 2. – Neutrino-produced muon flux from the two deep underground neutrino experiments of KRISHNASWAMY *et al.* and REINES *et al.* corrected for hypothetical isotropic flux I_T of low-energy pions and/or muons secondary to penetrating muon component is shown as a function of I_T at the depth of 8740 hg.

of I_T at 8740 hg. One can see that there is agreement between the two experiments only if $I_T \leqslant 5 \cdot 10^{-15}$ particles $\mathrm{cm^{-2}\,s^{-1}\,sr^{-1}}$ (at 8740 hg) in contrast with the value of $5 \cdot 10^{-14}$ predicted by KEUFFEL. If the mechanism proposed by KEUFFEL is taking place at all, it is happening with a cross-section which is approximately 10 times smaller than what we would require to explain the two puzzles.

We will now present an argument that is based on the spirit of the Keuffel model but independent of its details. The vertical muon intensities at the two sites, KGF and ERPM, differ by a factor ≈ 6 (see row 4 in the Table) and yet the results on the neutrino-produced muon flux are in agreement with each other (row 5). This agreement

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([13]) M. R. KRISHNASWAMY, M. G. K. MENON, V. S. NARASIMHAM, K. HINOTANI, N. ITO, S. MIYAKE, J. L. OSBORNE, A. J. PARSONS and A. W. WOLFENDALE: *Proc. Roy. Soc.* (1971), to be published.
([14]) F. REINES, W. R. KROPP, H. W. SOBEL, H. S. GURR, J. LATHROP, M. F. CROUCH, J. P. F. SELLSCHOP and B. S. MEYER: *Phys. Rev. D* (1971), in press.

TABLE I.

Site	KGF, India	ERPM, S. Africa
Reference	KRISHNASWAMY *et al.* [13]	REINES *et al.* [14]
Depth, hg of standard rock	7600	8740
Vertical intensity of atmospheric muons $I_v(\mu)$ in units of 10^{-13} cm^{-2} s^{-1} sr^{-1}	(1100 $\pm$200)	(187 $\pm$23)
Neutrino-produced muon flux $I_\nu(\mu)$ in units of 10^{-13} cm^{-2} s^{-1} sr^{-1}	(2.6$\pm$ 0.7)	(3.7$\pm$ 0.6)
Keuffel prediction for muon-produced particle flux I_T in units of 10^{-13} cm^{-2} s^{-1} sr^{-1} [6]	2.5	0.5
Corrected neutrino-produced muon flux 10^{-13} cm^{-2} s^{-1} sr^{-1}	$0.1^{+0.7}_{-0.1}$	(3.2$\pm$ 0.6)

puts a rather stringent limit to the magnitude of any isotropic flux I_T that is secondary to the vertical muon flux I_v. We have just shown that I_T cannot be more than $5\cdot10^{-15}$ particles cm^{-2} s^{-1} sr^{-1} at the ERPM site and, by scaling, not more than $2.5\cdot10^{-14}$ particles cm^{-2} s^{-1} sr^{-1} at the KGF site. The secondary low-energy particle fluxes thus constitute a very small fraction $r = I_T/I_v \leqslant 2.5\cdot10^{-4}$ of the vertical muon fluxes. This ratio is expected to depend very little on the depth; for, the shape of the differential muon energy spectrum at great depths is almost independent of depth. To resolve the discrepancy between the experiments on the angular distribution of muons of BERGESON *et al.* [1] and of KRISHNASWAMY *et al.* [2] and STOCKEL [3], one needs the existence of low-energy particle fluxes at large zenith angles of a magnitude $\sim (3\div5)\cdot10^{-2}$ of the vertical muon fluxes at the respective sites. Since we have shown here that the agreement between the two underground neutrino experiments [13,14] restricts this fraction r to less than $2.5\cdot10^{-4}$, one can say with confidence that only a very small fraction ($\leqslant0.008$) of the triggers at large zenith angles in the experiments of KRISHNASWAMY *et al.* [2] and STOCKEL [3] are caused by low-energy particles. This fraction is too small to explain the discrepancy. The puzzle, therefore, remains unresolved.

We conclude that, while the two puzzles described in the earlier part of this paper continue to be of great interest and in need of resolution, the model advanced by KEUFFEL to explain them seems to be unlikely.

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